

NtR (Q9W288 / CG6698, *Drosophila melanogaster*) — Function-Assignment Review

Focus: Does NtR *directly* have the assigned functions — acetylcholine-gated channel complex (GO:0005892), *contributes_to* acetylcholine-gated cation-selective channel activity (GO:0022848), calcium ion transport (GO:0006816), cholinergic/synaptic signaling (GO:0007271)?

Hypothesis slug: ntr-cholinergic-calcium-and-synaptic-functions **Source:** genes/DROME/NtR/NtR-ai-review.yaml (free-text)

Executive Judgment

Verdict: Partially supported / over-annotated at the specificity level (homology-only).

NtR is a *bona fide* member of the Cys-loop ligand-gated ion channel (LGIC) superfamily: it has a *bona fide* extracellular neurotransmitter-gated ion-channel ligand-binding domain (Pfam PF02931), one canonical Cys-loop signature (C337–x13–C351), and a four-transmembrane-helix C-terminal region (M1–M4) in the expected Cys-loop topology. At a **general** level, terms describing it as a **ligand-gated / cation-permeable ion channel of the Cys-loop family localized to the plasma membrane** are defensible.

However, the **specific** claims in the seed hypothesis — that the ligand is **acetylcholine**, that it is a **calcium** transporter, and that it functions in **cholinergic synaptic** transmission — are **not supported by any primary experimental evidence**. Every one of the 15 current GO annotations for Q9W288 is computational: the cholinergic/calcium/synaptic core is **IBA (phylogenetic inference, ECO:0000318)** propagated from the PANTHER subfamily, and the general channel terms are **IEA** from InterPro. There is **no IDA/IMP/IPI/EXP annotation, no electrophysiology, no heterologous-expression assay, and no mutant phenotype** for NtR.

Two independent lines of computational evidence **actively undercut the ACh-specific inheritance**: 1. NtR sits in its **own PANTHER family (PTHR36695)**, *not* the canonical insect nAChR family — consistent with Matthews et al. 2018

([P 30429615](#)

Extended Data 10d) placing the NTR branch **outside** canonical nicotinic AChRs. 2. NtR **lacks the loop-C vicinal cysteines** that define α -type ACh-binding subunits, and its **transmembrane/pore region is divergent** (it does not receive the canonical Neur_chan_memb pore fold; it gets a generic AcrB-like TM structural label). Cation-vs-anion selectivity and Ca^{2+} permeability therefore cannot be inferred from homology.

Bottom line for the curator: The Matthews phylogenetic placement means broad Cys-loop homology **cannot** be used to justify the ACh-, calcium-, and cholinergic-specific terms. These should be **generalized, flagged as non-core, or down-qualified** (retaining the `contributes_to` hedge where a subunit-level channel term is kept). The general LGIC/ion-channel/plasma-membrane terms may be retained. The hypothesis is **not refuted** (a synaptic role remains possible and is not disproven), but it is **not established** either.

Evidence Matrix

#	Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
1	UniProt Q9W288; QuickGO (GO_REF:0000033, :0000002, :0000108)	Computational (database)	Qualifies	Evidence basis of the annotations	All 15 GO annotations are IBA or IEA; zero experimental codes	<i>D. melanogaster</i> gene product record
2	Matthews et al. 2018, P 30429615 (ED Fig 10d; cited from seed)	Structural/ evolutionary	Qualifies / competing	Is NtR a canonical nAChR?	NTR branch placed outside canonical nicotinic AChRs	Aedes/insect Cys-loop phylogeny
3	InterPro/Pfam for Q9W288 (PF02931; PANTHER PTHR36695)	Structural/ evolutionary (computational)	Qualifies	Family placement	LBD confirms Cys-loop membership; but own PANTHER family PTHR36695, not nAChR family	Domain/ orthology analysis
4	This work — sequence analysis (Cys positions)	Structural/ evolutionary (computational)	Qualifies / refutes α- subunit	ACh-binding α -subunit determinants	One Cys-loop (C337–C351); no loop-C vicinal cysteines → non- α subunit	585-aa sequence
5	This work — Kyte-Doolittle hydropathy	Structural/ evolutionary (computational)	Supports (general) / qualifies (specific)	Channel topology	Signal peptide + LBD + 4 TM (M1– M4) → canonical Cys-loop architecture	585-aa sequence

#	Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
6	InterPro structural assignments (SSF63712 vs SSF82866; PF12248)	Structural/ evolutionary (computational)	Qualifies	Is the pore a canonical channel pore?	LBD gets Cys-loop folds; TM region gets generic AcrB TM fold, not Neur_chan_memb pore	Domain analysis
7	Jones, Brown & Sattelle 2007, P 17216517	Review	Qualifies	Existence of divergent subunits	"each insect possesses at least one highly divergent nAChR subunit"	Insect nAChR gene families
8	Dent 2006, P 16586016	Review/ evolutionary	Qualifies	α vs non- α clades	Invertebrate nAChRs split into α -type and non- α clades; large divergent nAChR-like groups exist	Cross-species Cys-loop phylogeny
9	FlyBase/mygene (gene 43935, FBgn0029147)	Database	Qualifies	Is there curated function?	No curated functional summary; only IBA/IEA GO	<i>D. melanogaster</i>

No primary functional assay, mutant phenotype, localization, or interaction study specific to NtR was found.

GO Curation Implications (leads — require curator verification)

GO ID	Term	Aspect	Current evidence	Lead action	Rationale
GO:0022848	ACh-gated cation-selective channel activity	MF	IBA	Generalize to e.g. GO:0022824 (transmitter-gated monoatomic cation channel) or GO:0005230, or keep with <code>contributes_to</code> + explicit low-confidence note	ACh ligand not assayed; subfamily outside canonical nAChRs
GO:0005892	acetylcholine-gated channel complex	CC	IBA	Generalize / flag	No evidence NtR co-assembles into an ACh receptor pentamer
GO:0006816	calcium ion transport	BP	IBA	Remove or generalize to GO:0034220 ion transmembrane transport	Ca ²⁺ permeability inherited from $\alpha 7$ -like members; NtR pore divergent, untested
GO:0007271	synaptic transmission, cholinergic	BP	IBA	Generalize / treat as non-core	Cholinergic identity unproven; no expression/synapse data
GO:0005230	extracellular ligand-gated ion channel activity	MF	IEA	Retain (general)	Supported by LBD homology; conservative

GO ID	Term	Aspect	Current evidence	Lead action	Rationale
					and defensible
GO:0034220	monoatomic ion transmembrane transport	BP	IBA/IEA	Retain (general)	Defensible from 4-TM channel topology
GO:0042391	regulation of membrane potential	BP	IBA	Retain (general)	Reasonable for an ion channel
GO:0005886	plasma membrane	CC	IBA	Retain	Consistent with signal peptide + 4-TM membrane protein
GO:0045202 / 0098794 / 0060079 / 0099565	synapse / postsynapse / EPSP / postsynaptic transmission	CC/BP	IEA (GOC auto)	Treat as non-core / flag	Auto-propagated from the cholinergic terms; no localization data

Preserve the `contributes_to` qualifier wherever a channel-*activity* term is retained: NtR would at most be one subunit contributing to a multimeric channel's activity, so `contributes_to` is the correct hedge and should not be dropped.

Do not default to "protein binding" — the LBD-based ligand-gated-ion-channel and ion-transport terms are more informative and are homology-supported at the general level.

Mechanistic Scope

- **Immediate molecular function actually tested by evidence:** membership in the Cys-loop LGIC superfamily via an extracellular ligand-binding domain and a 4-TM channel body. This supports a *general* "transmitter-gated ion channel subunit" identity.
- **Not established as direct activity:** (i) the identity of the gating ligand (acetylcholine vs GABA/histamine/glutamate/protons/orphan), (ii) cation vs anion selectivity, (iii) calcium permeability, (iv) participation in a specific ACh-receptor complex.
- **Downstream / inferred-only:** cholinergic synaptic transmission, EPSP generation, postsynaptic membrane-potential regulation — these are **propagated** from the (unproven) ACh assignment, not measured. They should be treated as pathway-context inferences, not core function.

Conflicts and Alternatives

1. **Phylogenetic placement conflict:** Matthews et al.

(P 30429615)

ED10d and the independent PANTHER family assignment (PTHR36695, not the nAChR family) both place NtR **outside** canonical nAChRs. IBA propagation of ACh-specific terms across a branch that the tree itself separates from nAChRs is internally inconsistent and is the central reason to down-qualify.

2. **Non- α subunit:** absence of loop-C vicinal cysteines means NtR cannot supply the principal (+) ACh-binding face even if it were in an ACh receptor — it could only ever be a structural/complementary subunit.
3. **Divergent pore:** the TM region does not match the canonical channel-pore fold, so cation selectivity and Ca^{2+} transport ($\alpha 7$ -like inheritance) are unsupported; NtR could be an anion channel or an orphan/atypical channel.
4. **Paralog/carry-over risk:** the ACh/calcium/synapse terms have the hallmark of homology carry-over from well-studied nAChR α -subunits onto a divergent "orphan" member — the classic over-annotation pattern the review is meant to catch.
5. **Alternative identity:** NtR resembles the "highly divergent" insect Cys-loop subunits (cf. pHCl, CG-series orphans) that are often **not** ACh-gated; several are pH-, GABA-, or ligand-unknown channels.

Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Gating ligand identity	Domains, phylogeny, sequence — no assay found	ACh-specific terms depend on it	Heterologous expression + agonist screen (ACh, GABA, Glu, His, protons)
Ion selectivity (cation vs anion; Ca^{2+})	Pore-fold divergence noted; M2 not confidently classifiable	Justifies "cation-selective" and "calcium transport"	Reversal-potential / ion-substitution electrophysiology
Subunit partners / complex	No interaction data	Justifies "acetylcholine-gated channel complex"	Co-IP / proteomics; co-expression functional reconstitution
Neuronal/ synaptic expression	No curated expression summary retrieved	Justifies synaptic BP/ CC terms	scRNA-seq brain atlas, reporter/antibody localization
Matthews ED10d specifics	Cited from seed; figure not retrievable here	Anchor for the "outside nAChR" claim	Curator to inspect P 30429615 ED Fig 10d and its PAINT nodes

Discriminating Tests

- Two-electrode voltage clamp / patch clamp** on NtR expressed in *Xenopus* oocytes or S2 cells, screening ACh vs GABA/Glu/histamine/protons — distinguishes ACh-gated from alternative-ligand or orphan channel.
- Ion-substitution reversal-potential experiments** and Ca^{2+} -imaging under agonist — tests cation selectivity and calcium permeability directly.
- Co-assembly assay** (tagged NtR + canonical $\text{D}\alpha/\text{D}\beta$ nAChR subunits) — tests membership in an ACh receptor complex.

4. **Single-cell brain/VNC expression atlas** query and endogenous tagging — tests neuronal/synaptic localization.
5. **Explicit inspection of the PANTHER PTHR36695 tree / PAINT annotations** — confirm which experimentally characterized node (if any) is the source of the IBA ACh terms; if no experimentally validated ACh-gated member exists in the subfamily, the IBA propagation is unsupported.

Curation Leads (require curator verification)

- **Action change:** Down-qualify or generalize the ACh-specific MF (GO:0022848), CC (GO:0005892), and BP (GO:0006816 calcium; GO:0007271 cholinergic) terms from confident to **homology-only/non-core**; retain general channel terms (GO:0005230, GO:0034220, GO:0042391, GO:0005886). **Keep** `contributes_to` on any retained channel-activity term.
- **Candidate reference to verify** —

P 30429615

(Matthews et al. 2018), Extended Data Fig 10d: verify that the NTR branch is drawn outside the canonical nicotinic AChR clade. *(Snippet must be verified by the curator against the figure; abstract text was not programmatically retrievable in this run.)*

- **Candidate reference** —

P 17216517

(Jones, Brown & Sattelle 2007): snippet to verify — *"each insect possesses at least one highly divergent nAChR subunit."*

- **Candidate reference** —

P 16586016

(Dent 2006): snippet to verify — *"There are two clades of invertebrate nAChRs, one of alpha-type subunits and one of non-alpha subunits..."*

- **Candidate replacement GO terms:** GO:0005230 (extracellular ligand-gated monoatomic ion channel activity) or GO:0022824 (transmitter-gated monoatomic cation channel activity) in place of the ACh-specific GO:0022848, pending assay.
- **Suggested questions for curator:** (1) Does the PTHR36695 subfamily contain any experimentally characterized ACh-gated member that legitimizes the IBA? (2) Is there any

transcriptomic evidence of neuronal expression? (3) Should the calcium-transport term be removed entirely given the divergent, untested pore?

- **Suggested experiments:** heterologous electrophysiology with a multi-ligand panel; ion-selectivity measurement; co-assembly test (as above).
-

Limitations of this review

- No primary functional literature exists for NtR; conclusions rest on sequence/domain/phylogeny analysis plus database evidence codes.
 - The Matthews ED10d figure and the PANTHER PAINT nodes were not programmatically retrievable in this run; the "outside canonical nAChR" placement is corroborated independently by the distinct PANTHER family (PTHR36695) but the exact figure should be curator-verified.
 - TM/pore selectivity classification (cation vs anion) was not confidently resolvable from sequence alone.
-

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery